



SYDNEY BOYS HIGH SCHOOL
MOORE PARK, SURRY HILLS

2009
TRIAL HIGHER SCHOOL
CERTIFICATE EXAMINATION

Mathematics Extension 2

General Instruction

- Reading Time – 5 Minutes
- Working time – 180 Minutes
- Write using black or blue pen.
Pencil may be used for diagrams.
- Board approved calculators maybe used.
- All necessary working should be shown in every question if full marks are to be awarded.
- Marks may **NOT** be awarded for messy or badly arranged work.
- Start each question in a separate answer booklet.
- Answer in simplest exact form unless otherwise instructed.

Total Marks – 120

- Attempt questions 1-8
- All questions are of equal value

Examiner: *C. Kourtesis*

This is an assessment task only and does not necessarily reflect the content or format of the Higher School Certificate

Question 1. (15 marks)

Marks

(a) Find: (i) $\int \frac{1}{\sqrt{x+8}} dx$ 3

(ii) $\int \frac{1}{x^2+9} dx$

(b) Use integration by parts to find $\int x \ln x dx$ 3

(c) Use completion of squares to find $\int \frac{dx}{\sqrt{6-x-x^2}}$ 2

(d) i) Find real numbers a , b and c such that $\frac{1}{x^2(2-x)} = \frac{ax+b}{x^2} + \frac{c}{2-x}$ 4

ii) Hence evaluate $\int_1^{1.5} \frac{dx}{x^2(2-x)}$

(e) Use the substitution $x = \tan y$ to show that $\int_0^1 \frac{dx}{(x^2+1)^2} = \frac{\pi+2}{8}$ 3

Question 2. (15 marks)

Marks

- (a) If k is a real number and $z = k - 2i$ express $\overline{(iz)}$ in the form $x + iy$ where x and y are real numbers. 2

- (b) Solve the equation 2

$$\bar{z} = 3z - 1$$

where $z = x + iy$ (x, y real)

- (c) On an Argand diagram shade the region specified by both the conditions 3

$$\operatorname{Im}(z) \leq 4 \text{ and } |z - 4 - 5i| \leq 3$$

- (d) If $\operatorname{cis} \theta = \cos \theta + i \sin \theta$ express 2

$$(4\operatorname{cis} \alpha)^2 (2\operatorname{cis} \beta)^3$$

in modulus-argument form.

- (e) i) Find the equation, in Cartesian form, of the locus of the point z if 4

$$\operatorname{Re}\left(\frac{z}{z+2}\right) = 0$$

ii) Sketch the locus of z satisfying the above.

- (f) If α and β are real show that $(\alpha + \beta i)^{2002} + (\beta - \alpha i)^{2002} = 0$. 2

Question 3. (15 marks)

Marks

(a) Consider the function

8

$$f(x) = \frac{x^3}{(1-x)^2}$$

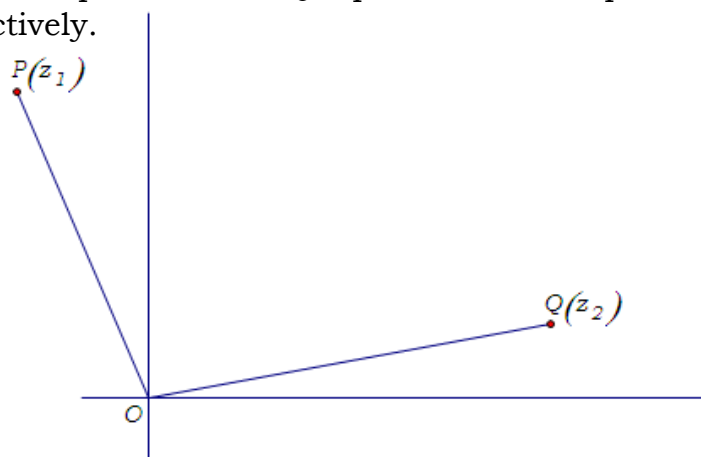
- i) Show that $f'(x) = \frac{x^2[3-x]}{(1-x)^3}$
- ii) Use the first derivative $f'(x)$ to determine the nature of the stationary points.
- iii) Write down the equations of any asymptotes.
- iv) Sketch the graph of $y = f(x)$ showing all essential features.

(b) i) Sketch the graphs of $y = \sin x$ and $y = \sqrt{\sin x}$ for $0 \leq x \leq \frac{\pi}{2}$ on the same diagram. 4

- ii) Hence show that $1 < \int_0^{\frac{\pi}{2}} \sqrt{\sin x} dx < \frac{\pi}{2}$

NOTE: You are NOT required to evaluate the integral $\int_0^{\frac{\pi}{2}} \sqrt{\sin x} dx$

(c) In the diagram below points P and Q represent the complex numbers z_1 and z_2 respectively. 3



- i) Copy the diagram in your examination booklet and indicate the point representing the complex number $z_1 + z_2$
- ii) If the length of PQ is $|z_1 - z_2|$ and $|z_1 - z_2| = |z_1 + z_2|$ what can be said about $\frac{z_2}{z_1}$

Question 4. (15 marks)

Marks

(a) The real cubic polynomial $ax^3 + 9x^2 + ax = 30$ has $-3+i$ as a root. 4

i) Show that $x^2 + 6x + 10$ is a quadratic factor of the cubic polynomial.

ii) Show that $a = 2$.

iii) Write down all the roots of the polynomial.

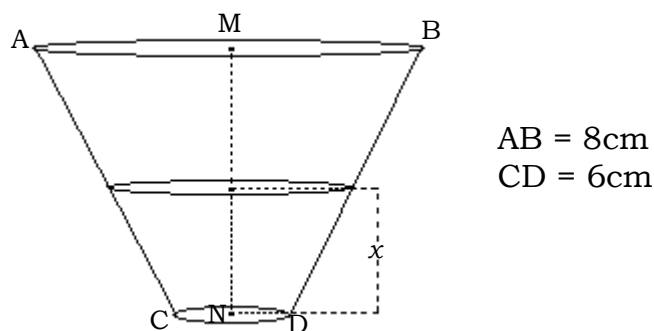
(b) Show that the polynomial $P(x) = nx^{n+1} - (n+1)x^n + 1$ is divisible by $(x-1)^2$ 2

(c) i) Sketch the graphs of $y = \frac{1}{x^2 + 1}$ and $y = \frac{1}{x^2 + 2}$ on the same set of axes. 4

ii) The area bounded by the two curves in (i) and the ordinates at $x = 0$ and $x = 2$ is rotated about the y-axis. Use the cylindrical shell method to show that the volume of the resulting solid is

$$\pi \ln \frac{5}{3}.$$

(d) A drinking glass is in the shape of a truncated cone, in which the internal diameter of the top and bottom are 8cm and 6cm respectively. 5



i) If the internal height of the glass, MN, is 10cm show that the area of the cross-section x cm above the base is

$$\pi \left(3 + \frac{x}{10} \right)^2 \text{ cm}^2.$$

ii) Hence find by integration, the volume of liquid the glass can hold (answer to the nearest mL).

Question 5. (15 marks)**Marks**

The equation of an ellipse E is given by $\frac{x^2}{9} + \frac{y^2}{5} = 1$

- | | |
|--|---|
| i) Find the eccentricity of E | 1 |
| ii) Write down the | 3 |
| α) coordinates of the foci | |
| β) equations of the directrices | |
| γ) equation of the major auxiliary circle A. | |
| iii) Draw a neat sketch of E showing clearly the features in part ii) | 2 |
| iv) A line parallel to the positive y-axis meets the x-axis at N and the curves E, A at P and Q respectively. If N has coordinates $(3\cos\theta, 0)$ find the coordinates of P and Q. [P and Q are in the first quadrant] | 2 |
| v) Show that the equations of the tangents at P and Q are $\sqrt{5}x\cos\theta + 3y\sin\theta = 3\sqrt{5}$ and $x\cos\theta + y\sin\theta = 3$ respectively. | 4 |
| vi) Show that the point of intersection R of these tangents lies on the major axis of E produced. | 1 |
| vii) Prove that $ON \cdot OR$ is independent of the position of P and Q on the curves. | 2 |

Question 6. (15 marks)

Marks

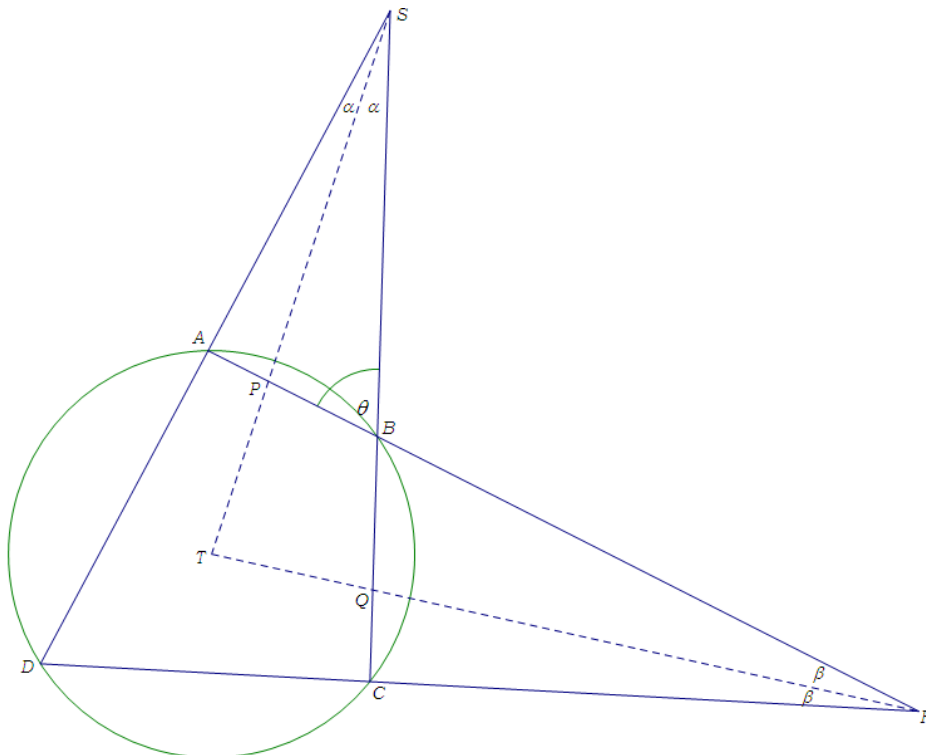
- (a) i) A particle of mass m falls vertically from rest, from a point o , in a medium whose resistance is mkv , where k is a positive constant and v its velocity after t seconds. 4

$$\text{Show that } v = \frac{g}{k}(1 - e^{-kt})$$

- ii) An equal particle is projected vertically upwards with initial velocity U in the same medium. [The particle is released simultaneously with the first particle]. 4

Show that the velocity of the first particle when the second particle is momentarily at rest is given by $\frac{VU}{V+U}$ where V is the terminal velocity of the first particle.

- (b) 7



$ABCD$ is a cyclic quadrilateral.

The sides AB and CD produced intersect at R and the sides CB and DA produced intersect at S . ST and RT intersect AR and CS at P and Q respectively.

The bisectors of $\hat{CSD}$ and $\hat{ARD}$ meet at T .

Let $\hat{AST} = \hat{BST} = \alpha$ and $\hat{ART} = \hat{DRT} = \beta$ and $\hat{ABS} = \theta$.

- i) Show that $\hat{TPB} + \hat{TQB} = \alpha + \beta + 2\theta$
- ii) Prove that ST is perpendicular to RT .

Question 7. (15 marks)

Marks

- (a) Given that $\tan 5\theta = \frac{t^5 - 10t^3 + 5t}{5t^4 - 10t^2 + 1}$, where $t = \tan \theta$ [Do not prove this] 5

i) Solve the equation $\tan 5\theta = 0$ for $0 \leq \theta \leq \pi$

ii) Hence prove that

$\alpha) \tan \frac{\pi}{5} \tan \frac{2\pi}{5} = \sqrt{5}$

$\beta) \tan^2 \frac{\pi}{5} + \tan^2 \frac{2\pi}{5} = 10$

- (b) i) Show that $\int x \tan^{-1} x \, dx = \frac{1}{2}(x^2 + 1) \tan^{-1} x - \frac{1}{2}x + c$ 4

ii) If $u_n = \int_0^1 x^n \tan^{-1} x \, dx$ for $n \geq 2$ show that

$$u_n = \frac{\pi}{2(n+1)} - \frac{1}{n(n+1)} - \frac{n-1}{n+1} u_{n-2}$$

- (c) Show that the number of ways in which $2n$ persons may be seated at two round tables, n persons being seated at each is 2

$$\frac{(2n)!}{n^2}$$

- (d) i) There are 6 persons from whom a game of tennis is to be made up, two on each side. How many different matches can be arranged if a change in either pair gives a different match? 4

ii) How many different matches are possible if two particular persons are to both play in the match?

Question 8. (15 marks)

Marks

(a) Suppose a, b, c and d are positive real numbers.

5

i) Prove that $\frac{a}{b} + \frac{b}{a} \geq 2$.

ii) Deduce that $\frac{a+b+c}{d} + \frac{b+c+d}{a} + \frac{c+d+a}{b} + \frac{d+a+b}{c} \geq 12$.

iii) Hence prove that if $a + b + c + d = 1$, then:

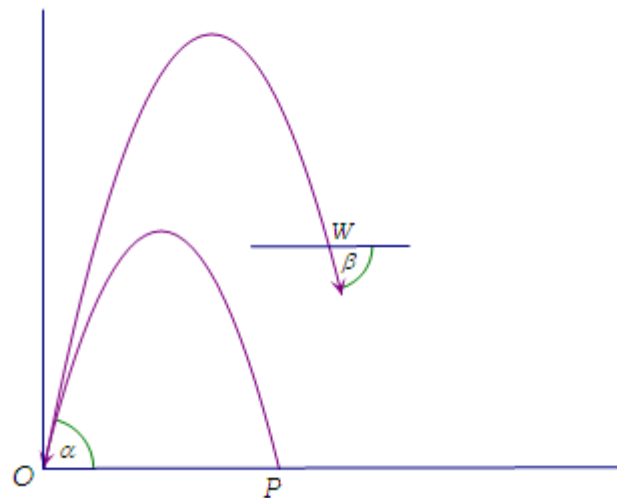
$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d} \geq 16.$$

(b) Two stones are thrown simultaneously from the same point O in the same direction and with the same non-zero angle of projection α , but with different velocities U and V ($U < V$).

6

The slower stone hits the ground at a point P on the same level as the point of projection.

At that instant the faster stone is at a point W on its downward path, making an angle β with the horizontal.



i) Show that $V(\tan \alpha + \tan \beta) = 2U \tan \alpha$

ii) Deduce that if $\beta = \frac{1}{2}\alpha$ then $U < \frac{3}{4}V$

(c) i) Show by graphical means that

4

$$\ln ex > e^{-x} \text{ for } x \geq 1$$

ii) Hence, or otherwise, show that

$$\ln(n!e^n) > e^{-n} \left(\frac{e^n - 1}{e - 1} \right)$$

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax,$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

$$\begin{aligned}
 1) a) i) \int \frac{1}{\sqrt{x+8}} dx &= \int (x+8)^{-\frac{1}{2}} dx \\
 &= \frac{(x+8)^{\frac{1}{2}}}{\frac{1}{2} \cdot 1} + C \\
 &= 2\sqrt{x+8} + C
 \end{aligned}$$

$$ii) \int \frac{1}{x^2+9} dx = \frac{1}{3} \tan^{-1}\left(\frac{x}{3}\right) + C$$

$a=3$

$$\begin{aligned}
 b) \int x \ln x dx &= \frac{x^2}{2} \ln x - \frac{1}{2} \int x dx \\
 &= \frac{x^2}{2} \ln x - \frac{x^2}{4} + C
 \end{aligned}$$

$$\begin{array}{lcl}
 u = \ln x & \rightarrow & v' = x \\
 u' = \frac{1}{x} & \leftarrow & v = \frac{x^2}{2}
 \end{array}$$

$$c) \int \frac{dx}{\sqrt{6-x-x^2}} = \int \frac{dx}{\sqrt{-(x^2+x+\frac{1}{4})+\frac{25}{4}}}$$

$$= \int \frac{dx}{\sqrt{\frac{25}{4} - (x+\frac{1}{2})^2}}$$

$$a = \frac{5}{2}$$

OR make a substitution

$$= \sin^{-1}\left(\frac{(x+\frac{1}{2})}{(\frac{5}{2})}\right) + C$$

$$= \sin^{-1}\left(\frac{2x+1}{5}\right) + C$$

$$d) i) \frac{1}{x^2(2-x)} \equiv \frac{ax+b}{x^2} + \frac{c}{2-x}$$

$$1 \equiv (ax+b)(2-x) + cx^2$$

$$\text{let } x=2$$

$$1 = c(2)^2$$

$$c = \frac{1}{4}$$

$$\text{let } x=0$$

$$1 = b \cdot 2$$

$$b = \frac{1}{2}$$

equate coefficients of x^2

$$0 = -a + c$$

$$a = c$$

$$\therefore a = \frac{1}{4}$$

$$a = \frac{1}{4}, b = \frac{1}{2}, c = \frac{1}{4}$$

$$\text{ii) } \int_1^{1.5} \frac{dx}{x^2(2-x)} = \int_1^{1.5} \left(\frac{\frac{1}{4}x + \frac{1}{2}}{x^2} + \frac{\frac{1}{4}}{2-x} \right) dx$$

$$= \int_1^{1.5} \left(\frac{1}{4} \cdot \frac{1}{x} + \frac{1}{2} x^{-2} - \frac{1}{4} \cdot \frac{-1}{2-x} \right) dx$$

$$= \left[\frac{1}{4} \ln x - \frac{1}{2} x^{-1} - \frac{1}{4} \ln(2-x) \right]_1^{1.5}$$

$$= \left[-\frac{1}{2x} + \frac{1}{4} \ln \left(\frac{x}{2-x} \right) \right]_1^{1.5}$$

$$= -\frac{1}{2(1.5)} + \frac{1}{4} \ln \left(\frac{1.5}{2-1.5} \right) - \left(-\frac{1}{2(1)} + \frac{1}{4} \ln \left(\frac{1}{2-1} \right) \right)$$

$$= -\frac{1}{3} + \frac{1}{4} \ln 3 + \frac{1}{2}$$

$$= \frac{1}{4} \ln 3 + \frac{1}{6}$$

$$\text{e) } \int_0^1 \frac{dx}{(x^2+1)^2}$$

$$x = \tan y$$

$$\frac{dx}{dy} = \sec^2 y$$

$$dx = \sec^2 y \cdot dy$$

$$\text{when } x=1 \quad y = \frac{\pi}{4}$$

$$x=0 \quad y=0$$

$$\int_0^1 \frac{dx}{(x^2+1)^2} = \int_0^{\frac{\pi}{4}} \frac{\sec^2 y \, dy}{(\tan^2 y + 1)^2}$$

$$= \int_0^{\frac{\pi}{4}} \frac{\sec^2 y}{(\sec^2 y)^2} \, dy$$

$$= \int_0^{\frac{\pi}{4}} \frac{1}{\sec^2 y} \, dy$$

$$= \int_0^{\frac{\pi}{4}} \cos^2 y \, dy$$

$$= \int_0^{\frac{\pi}{4}} \left(\frac{1}{2} + \frac{1}{2} \cos 2y \right) dy$$

$$= \left[\frac{1}{2}y + \frac{1}{4} \sin 2y \right]_0^{\frac{\pi}{4}}$$

$$= \frac{1}{2} \left(\frac{\pi}{4} \right) + \frac{1}{4} \sin 2 \left(\frac{\pi}{4} \right) - \left(\frac{1}{2}(0) + \frac{1}{4} \sin 2(0) \right)$$

$$= \frac{\pi}{8} + \frac{1}{4}(1)$$

$$= \frac{\pi + 2}{8}$$

Question 2

(a) $z = k - 2i$

$$\begin{aligned}\overline{iz} &= \overline{i(k-2i)} \\ &= \overline{ik+2} \\ &= 2 - ik\end{aligned}$$

$$x = 2, y = -k$$

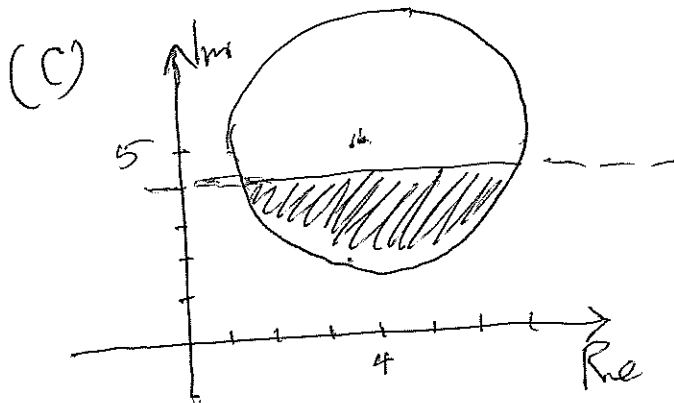
(b) $\bar{z} = 3z - 1$

$$x - iy = 3(x + iy) - 1$$

$$0 = 2x - 1 + 4iy$$

$$\begin{aligned}\text{Real: } 2x - 1 &= 0 & \text{Imag: } 4y &= 0 \\ x &= \frac{1}{2} & y &= 0\end{aligned}$$

Solution $z = \frac{1}{2}$



$$\begin{aligned}(d) & (4 \operatorname{cis} \alpha)^2 (2 \operatorname{cis} \beta)^3 \\ &= 16 \operatorname{cis} 2\alpha \cdot 8 \operatorname{cis} 3\beta \\ &= 128 \operatorname{cis} (2\alpha + 3\beta)\end{aligned}$$

(f) Let $z = \alpha + i\beta$

$$\text{Now } -iz = \beta - i\alpha$$

$$\begin{aligned}\text{Thus } (\alpha + i\beta)^{2002} + (\beta - i\alpha)^{2002} \\ &= z^{2002} + (-iz)^{2002} \\ &= z^{2002} + (-1)^{2002} \cdot z^{2002} \\ &= z^{2002} - z^{2002} \\ &= 0.\end{aligned}$$

CP3

(a). $f(x) = \frac{x^3}{(1-x)^2}$

(i)
$$f'(x) = \frac{(1-x)^2 \cdot 3x^2 - x^3 \cdot 2(1-x)^{-1} \cdot (-1)}{(1-x)^4}$$
$$= \frac{3x^2(1-x) + 2x^3}{(1-x)^3}$$
$$= \frac{3x^2 - 3x^3 + 2x^3}{(1-x)^3}$$
$$= \frac{3x^2 - x^3}{(1-x)^3}$$
$$= \frac{x^2(3-x)}{(1-x)^3}$$

(ii) Let $f'(x) = 0$
ie. $x = 0, 3$.
 $\therefore y = 0, \frac{27}{4}$

Test $(3, \frac{27}{4})$

Test $(0, 0)$

x	-1	0	$\frac{1}{2}$
y'	$\frac{1}{2}$	0	5

$\therefore$ STATIONARY INFLECTION

x	2	3	4
y'	-4	0	$\frac{16}{27}$

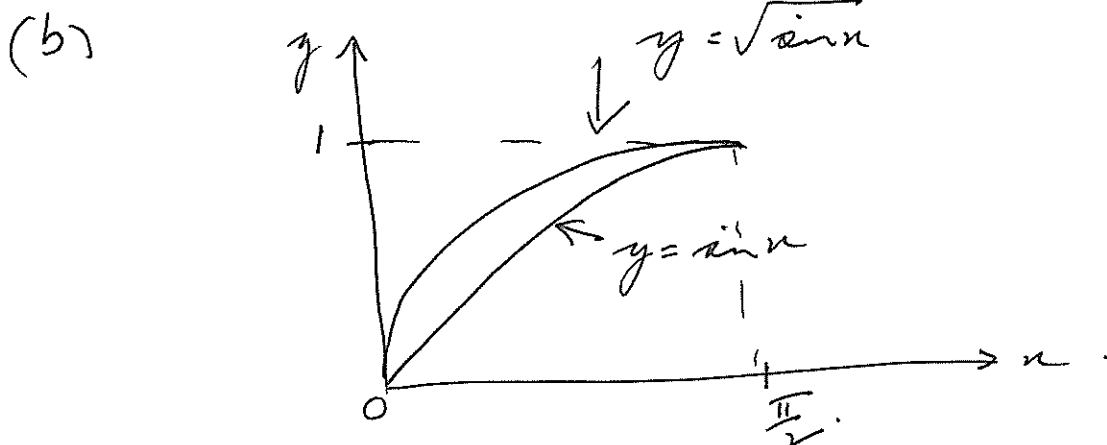
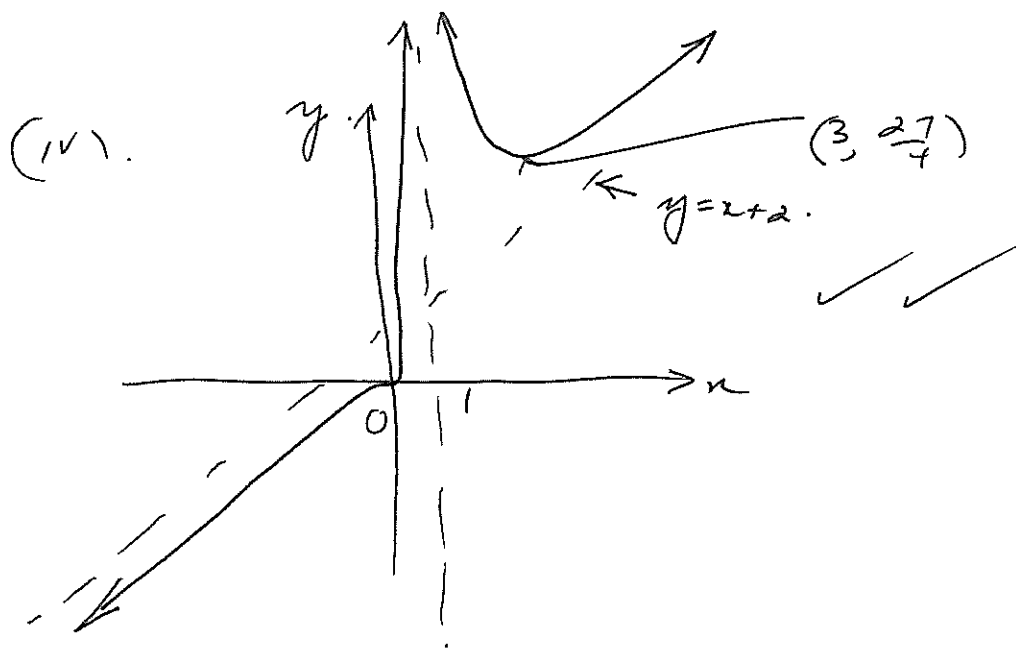
REL. MIN. TURNING PT.

(iii) Vertical Asymptote at $x = 1$

Now $\frac{x^3}{x^2 - vx + 1} = \frac{x(x^2 - vx + 1) + 2(x^2 - vx + 1) + 3x - 2}{x^2 - vx + 1}$

$$= x + 2 + \frac{3x - 2}{x^2 - vx + 1} \rightarrow x + 2 \text{ as } x \rightarrow \infty.$$

$\therefore y = x + 2$ is an oblique asymptote ✓



now $\int_0^{\frac{\pi}{2}} \sin x \, dx < \int_0^{\frac{\pi}{2}} \sqrt{\sin x} \, dx < \frac{\pi}{2} \times 1.$

$$[-\cos x]_0^{\frac{\pi}{2}} < \int_0^{\frac{\pi}{2}} \sqrt{\sin x} \, dx < \frac{\pi}{2}.$$

$$0 - -1 < \int_0^{\frac{\pi}{2}} \sqrt{\sin x} \, dx < \frac{\pi}{2}$$

$$1 < \int_0^{\frac{\pi}{2}} \sqrt{\sin x} \, dx < \frac{\pi}{2}.$$

(contd)

(b)

For $P(x)$ to be divisible by $(x-1)^2$

then $P(x)=0$ must have a multiple root of degree 2, of value 1.

$$\begin{aligned}\text{now } P(1) &= n - (n+1) + 1 \\ &= n - n - 1 + 1 \\ &= 0.\end{aligned}$$

$$P'(x) = n(n+1)x^n - n(n+1)x^{n-1}$$

$$\begin{aligned}\therefore P'(1) &= n(n+1) - n(n+1) \\ &= 0.\end{aligned}$$

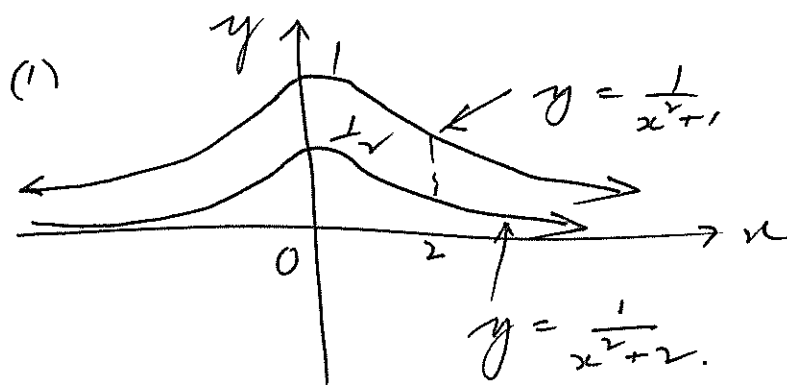
$$\therefore P(1) = P'(1) = 0$$

$\therefore$ by the multiple root theorem

$x=1$ is a double root.

$\therefore (x-1)^2$ is a factor.

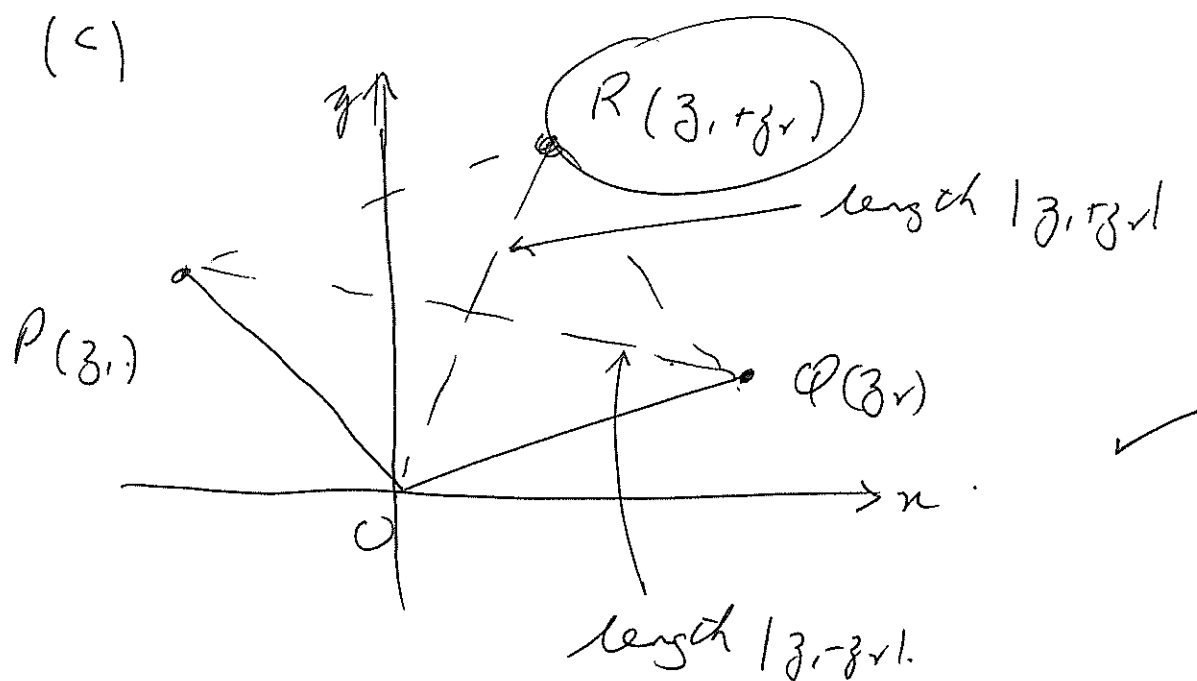
(c)



(ii)

$$\delta V = 2\pi x \left(\frac{1}{x^2+1} - \frac{1}{x^2+2} \right) \delta x$$

$$V = \lim_{\delta x \rightarrow 0} \sum_{x=0}^{\infty} 2\pi x \left(\frac{1}{x^2+1} - \frac{1}{x^2+2} \right) \delta x$$



now the diagonals of the parallelogram
are equal $\therefore$ a ~~rectangle~~ rectangle.

$$\therefore z_1 = k i z_2 \quad \text{or} \quad z_2 = -k i z_1 \quad \checkmark$$

$$\frac{z_2}{z_1} = -k i \quad \checkmark$$

i IMAGINARY

C(continued)

$$\begin{aligned}
 &= 2\pi \int_0^2 \left(\frac{x}{x^2+1} - \frac{x}{x^2+4} \right) dx \\
 &= \pi \left[\ln(x^2+1) - \ln(x^2+4) \right]_0^2 \\
 &= \pi \left[\ln 5 - \ln 6 - \ln 1 + \ln 2 \right] \\
 &= \pi \ln \frac{10}{6} \\
 &= \left(\pi \ln \frac{5}{3} \text{ m}^2 \right)
 \end{aligned}$$

(d). (i) Let the radius of the cross section be r .

$$\begin{aligned}
 \therefore \text{by similarity } \frac{r-3}{x} &= \frac{1}{10} \\
 r &= 3 + \frac{x}{10}
 \end{aligned}$$

$\therefore$ Area of the cross-section is $\pi \left(3 + \frac{x}{10} \right)^2$

$$\begin{aligned}
 \therefore (ii) \quad V &= \lim_{\delta x \rightarrow 0} \sum_{x=0}^{10} \pi \left(3 + \frac{x}{10} \right)^2 \delta x \\
 &= \pi \int_0^{10} \left(3 + \frac{x}{10} \right)^2 dx \\
 &= \frac{10\pi}{3} \left[\left(3 + \frac{x}{10} \right)^3 \right]_0^{10} \\
 &= \frac{10\pi}{3} [4^3 - 3^3] \\
 &= \frac{370\pi}{3} \text{ cc.} \\
 &\approx \boxed{387 \text{ ml.}}
 \end{aligned}$$

Q4.

(a) (i) given $ax^3 + 9x^2 + ax - 30 = 0$

with real co-efficients, has a root $-3+i$, it also has $-3-i$ as a root, by the conjugate root theorem.

$$\therefore x^2 - (-3+i + -3-i)x + (-3+i)(-3-i)$$

is a factor.

$$\text{i.e. } \boxed{x^2 + 6x + 10}$$

(ii) now clearly $ax^3 + 9x^2 + ax - 30 \equiv (x^2 + 6x + 10)(ax$

now co-eff of x

$$\text{LHS} = a \quad \text{RHS} = 10a - 18$$

$$\therefore 10a - 18 = a$$

$$9a = 18$$

$$\boxed{a = 2}$$

$$(iii) \sum \alpha_i = -\frac{9}{a} \quad \therefore -3+i + -3-i + \alpha = -\frac{9}{2}$$
$$-6 + \alpha = -\frac{9}{2}$$

$$\alpha = \frac{3}{2}$$

$$\therefore \text{roots are } \boxed{-3 \pm i, \frac{3}{2}}$$

(b) (see next page)

[15]

$$\frac{x^2}{9} + \frac{y^2}{5} = 1$$

(i) $b^2 = a^2(1-e^2)$

[1] $5 = 9(1-e^2)$
 $e^2 = 4/9 \Rightarrow e = \frac{2}{3}$

(ii) a) $(\pm ae, 0)$

$\therefore (\pm 2, 0)$

[3] b) $x = \pm \frac{a}{e}$
 $= \pm \frac{9}{2}$

c) $x^2 + y^2 = 9$

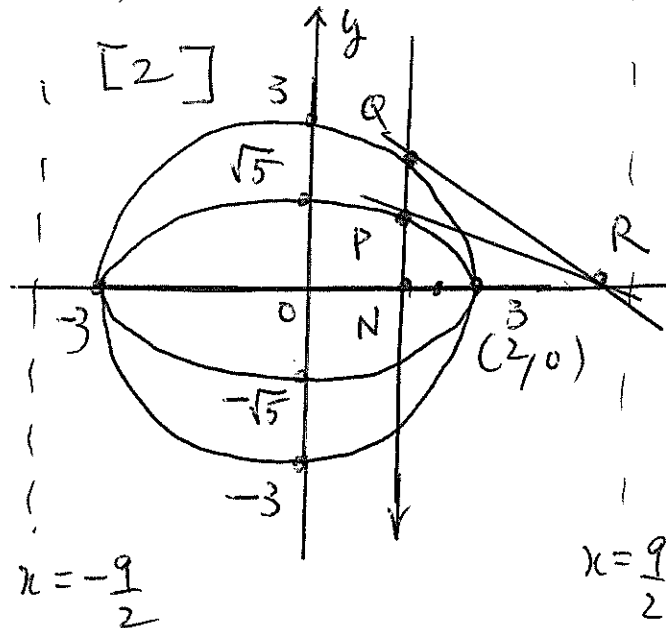
(iv) $N = (3 \cos \theta, 0)$

[2] $P (3 \cos \theta, \sqrt{5} \sin \theta)$

$Q (3 \cos \theta, 3 \sin \theta)$

Solution to Question (5)

(iii)



(v) At $P (3 \cos \theta, \sqrt{5} \sin \theta)$

$$\frac{2x}{9} + \frac{2y}{5} \frac{dy}{dx} = 0$$

$$\therefore \frac{dy}{dx} = -\frac{5x}{9y}$$

$$= -\frac{15 \cos \theta}{9 \sqrt{5} \sin \theta}$$

$$= -\frac{\sqrt{5} \cos \theta}{3 \sin \theta}$$

(1) $\therefore y - \sqrt{5} \sin \theta = -\frac{\sqrt{5} \cos \theta}{3 \sin \theta} (x - 3 \cos \theta)$

$$(\sin \theta) 3y - 3\sqrt{5} \sin^2 \theta = -\sqrt{5} x \cos \theta + 3\sqrt{5} \cos \theta$$

$$\therefore \sqrt{5} x \cos \theta + 3y \sin \theta = 3\sqrt{5}$$

$$2x + 2y \frac{dy}{dx} = 0$$

$$\therefore \frac{dy}{dx} = -\frac{\cos \theta}{\sin \theta}$$

$$y - 3 \sin \theta = \frac{-\cos \theta}{\sin \theta} (x - 3 \cos \theta)$$

$$\therefore x \cos \theta + y \sin \theta = 3$$

(vi) Solve (1) & (2) $y = 0$

$$\therefore x = \frac{3}{\cos \theta} = 3 \sec \theta$$

$$\therefore R (3 \sec \theta, 0)$$

(vii) $|ON| = |3 \cos \theta|$
 $|OR| = |3 \sec \theta|$

$$\therefore |ON| \cdot |OR| = |3 \cos \theta \times \frac{3}{\cos \theta}|$$

$$= 9$$

2009 Mathematics Extension 2 Trial HSC: Questions 7 & 8 solutions

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7. (a) Given that $\tan 5\theta = \frac{t^5 - 10t^3 + 5t}{5t^4 - 10t^2 + 1}$, where $t = \tan \theta$ [Do not prove this],

i) Solve the equation $\tan 5\theta = 0$ for $0 \leq \theta \leq \pi$.

Solution: $\tan 5\theta = 0,$
 $5\theta = 0 + n\pi, n = 0, 1, 2, 3, \dots$
 $\theta = \frac{n\pi}{5},$
 $= 0, \frac{\pi}{5}, \frac{2\pi}{5}, \frac{3\pi}{5}, \frac{4\pi}{5}, \pi.$

ii) Hence prove that

$\alpha) \tan \frac{\pi}{5} \tan \frac{2\pi}{5} = \sqrt{5},$

Solution: Method 1—

$$\begin{aligned} t^5 - 10t^3 + 5t &= 0, \\ t(t^4 - 10t^2 + 5) &= 0, \\ t = 0 \text{ or } t^2 &= \frac{10 \pm \sqrt{100 - 20}}{2}, \\ &= 5 \pm 2\sqrt{5}, \\ \text{So } t &= \pm(5 \pm 2\sqrt{5}). \\ \tan \frac{\pi}{5} &= \sqrt{5 - 2\sqrt{5}}, & \tan \frac{3\pi}{5} &= -\sqrt{5 + 2\sqrt{5}}, \\ \tan \frac{2\pi}{5} &= \sqrt{5 + 2\sqrt{5}}, & \tan \frac{4\pi}{5} &= -\sqrt{5 - 2\sqrt{5}}, \\ \therefore \tan \frac{\pi}{5} \tan \frac{2\pi}{5} &= \sqrt{25 - 20}, \\ &= \sqrt{5}. \end{aligned}$$

Solution: Method 2—

$$\begin{aligned} t^5 - 10t^3 + 5t &= 0, \\ t(t^4 - 10t^2 + 5) &= 0, \\ t = 0 \text{ or } t^4 - 10t^2 + 5 &= 0. \\ \text{i.e. } \tan \frac{\pi}{5} \times \tan \frac{2\pi}{5} \times \tan \frac{3\pi}{5} \times \tan \frac{4\pi}{5} &= 5, \text{ (product of roots)} \\ \tan \frac{\pi}{5} \times \tan \frac{2\pi}{5} \times \left(-\tan \frac{2\pi}{5}\right) \times \left(-\tan \frac{\pi}{5}\right) &= 5, \\ \text{i.e. } \tan^2 \frac{\pi}{5} \times \tan^2 \frac{2\pi}{5} &= 5, \\ \text{Hence } \tan \frac{\pi}{5} \tan \frac{2\pi}{5} &= \sqrt{5}. \\ \text{(Positive as both } \frac{\pi}{5}, \text{ and } \frac{2\pi}{5} \text{ are in the 1}^{\text{st}} \text{ quadrant).} \end{aligned}$$

$\beta) \tan^2 \frac{\pi}{5} + \tan^2 \frac{2\pi}{5} = 10.$

Solution: Method 1—

$$\begin{aligned} \tan^2 \frac{\pi}{5} + \tan^2 \frac{2\pi}{5} &= 5 - 2\sqrt{5} + 5 + 2\sqrt{5}, \\ &= 10. \end{aligned}$$

Solution: Method 2— (taking roots 2 at a time)

$$\begin{aligned}
 -10 &= \tan \frac{\pi}{5} \tan \frac{2\pi}{5} + \tan \frac{\pi}{5} \tan \frac{3\pi}{5} + \tan \frac{\pi}{5} \tan \frac{4\pi}{5} + \tan \frac{2\pi}{5} \tan \frac{3\pi}{5} \\
 &\quad + \tan \frac{2\pi}{5} \tan \frac{4\pi}{5} + \tan \frac{3\pi}{5} \tan \frac{4\pi}{5}, \\
 &= \tan \frac{\pi}{5} \tan \frac{2\pi}{5} + \tan \frac{\pi}{5} \left(-\tan \frac{2\pi}{5} \right) + \tan \frac{\pi}{5} \left(-\tan \frac{\pi}{5} \right) + \tan \frac{2\pi}{5} \left(-\tan \frac{2\pi}{5} \right) \\
 &\quad + \tan \frac{2\pi}{5} \left(-\tan \frac{\pi}{5} \right) + \left(-\tan \frac{2\pi}{5} \right) \left(-\tan \frac{\pi}{5} \right), \\
 &= -\tan^2 \frac{\pi}{5} - \tan^2 \frac{2\pi}{5}, \\
 10 &= \tan^2 \frac{\pi}{5} + \tan^2 \frac{2\pi}{5}.
 \end{aligned}$$

(b) i) Show that $\int x \tan^{-1} x \, dx = \frac{1}{2}(x^2 + 1) \tan^{-1} x - \frac{1}{2}x + c$.

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Solution: $I = \frac{x^2}{2} \tan^{-1} x - \frac{1}{2} \int \frac{1+x^2-1}{1+x^2} dx, \quad u = \tan^{-1} x, \quad v' = x \, dx,$
 $= \frac{x^2}{2} \tan^{-1} x - \frac{x}{2} + \frac{\tan^{-1} x}{2} + c, \quad u' = \frac{dx}{1+x^2}, \quad v = \frac{x^2}{2}.$
 $= \frac{1}{2}(x^2 + 1) \tan^{-1} x - \frac{x}{2} + c.$

ii) If $u_n = \int_0^1 x^n \tan^{-1} x \, dx$ for $n \geq 2$, show that

$$u_n = \frac{\pi}{2(n+1)} - \frac{1}{n(n+1)} - \frac{n-1}{n+1} u_{n-2}.$$

Solution: Method 1—

$$\begin{aligned}
 u_n &= \int_0^1 x^n \tan^{-1} x \, dx, & u &= x^{n-1}, \\
 &= \left[\frac{x^{n-1}(x^2+1) \tan^{-1} x}{2} - \frac{x^n}{2} \right]_0^1 & u' &= (n-1)x^{n-2} dx, \\
 &\quad - \frac{n-1}{2} \int_0^1 (x^2+1)x^{n-2} \tan^{-1} x \, dx & v' &= x \tan^{-1} x \, dx, \\
 &\quad + \frac{n-1}{2} \int_0^1 x^{n-1} dx, & v &= \frac{x^2+1}{2} \tan^{-1} x - \frac{x}{2}. \\
 &= \frac{\pi}{4} - \frac{1}{2} - \frac{n-1}{2} \int_0^1 x^n \tan^{-1} x \, dx \\
 &\quad - \frac{n-1}{2} \int_0^1 x^{n-2} \tan^{-1} x \, dx \\
 &\quad + \frac{n-1}{2} \left[\frac{x^n}{n} \right]_0^1, \\
 \left(1 + \frac{n-1}{2}\right) u_n &= \frac{\pi}{4} - \frac{n}{2n} - \frac{n-1}{2} u_{n-2} + \frac{n-1}{2} \cdot \frac{1}{n}, \\
 \frac{n+1}{2} u_n &= \frac{\pi}{4} - \frac{n-1}{2} u_{n-2} - \frac{1}{2n}, \\
 u_n &= \frac{\pi}{2(n+1)} - \frac{1}{n(n+1)} - \frac{n-1}{n+1} u_{n-2}.
 \end{aligned}$$

Solution: Method 2—

$$\begin{aligned}
 u_n &= \int_0^1 x^n \tan^{-1} x \, dx, & u &= \tan^{-1} x, \\
 &= \left[\frac{x^{n+1} \tan^{-1} x}{n+1} \right]_0^1 - \frac{1}{n+1} \int_0^1 \frac{x^{n+1}}{1+x^2} dx, & u' &= \frac{1}{1+x^2} dx, \\
 &= \frac{\pi}{4(n+1)} - \frac{1}{n+1} \int_0^1 \frac{(x^2+1)x^{n-1} - x^{n-1}}{1+x^2} dx, & v' &= x^n dx, \\
 &= \frac{\pi}{4(n+1)} - \frac{1}{n+1} \int_0^1 x^{n-1} dx + \frac{1}{n+1} \int_0^1 \frac{x^{n-1}}{1+x^2} dx, & v &= \frac{x^{n+1}}{n+1}. \\
 &= \frac{\pi}{4(n+1)} - \frac{1}{n+1} \left[\frac{x^n}{n} \right]_0^1 + \frac{1}{n+1} [x^{n-1} \tan^{-1} x]_0^1 \\
 &\quad - \frac{n-1}{n+1} \int_0^1 x^{n-2} \tan^{-1} x \, dx, & u &= x^{n-1}, \\
 &= \frac{\pi}{4(n+1)} - \frac{1}{n(n+1)} + \frac{\pi}{4(n+1)} - \left(\frac{n-1}{n+1} \right) u_{n-2}, & u' &= (n-1)x^{n-2} dx, \\
 &= \frac{\pi}{2(n+1)} - \frac{1}{n(n+1)} - \frac{n-1}{n+1} u_{n-2}, & v' &= \frac{1}{1+x^2} dx, \\
 & & v &= \tan^{-1} x.
 \end{aligned}$$

- (c) Show that the number of ways in which $2n$ persons may be seated at two round tables, n persons being seated at each is

$$\frac{(2n)!}{n^2}.$$

Solution: Ways of choosing people for one table is ${}^{2n}C_n = \frac{(2n)!}{(2n-n)!n!}$.

Ways of arranging each table is $(n-1)!$

$$\begin{aligned}
 \therefore \text{Total ways} &= \frac{(2n)!}{n!n!} \cdot (n-1)!(n-1)! \\
 &= \frac{(2n)!}{n^2}.
 \end{aligned}$$

- (d) i) There are 6 persons from whom a game of tennis is to be made up, two on each side. How many different matches can be arranged if a change in either pair gives a different match?

Solution: Ways of choosing 1st pair = 6C_2 ,

ways of choosing 2nd pair = 4C_2 .

But pair order not important,

$$\begin{aligned}
 \therefore \text{Number of matches} &= \frac{6!}{4!2!} \cdot \frac{4!}{2!2!} \cdot \frac{1}{2}, \\
 &= 45.
 \end{aligned}$$

- ii) How many different matches are possible if two particular persons are to both play in the match?

Solution: If the two are on the same team,

we only need to choose the other team: ${}^4C_2 = 6$.

If the two are on opposing teams,

(4 ways to get one partner) $\times$ (3 ways to get the other) = 12.

$\therefore$ Number of matches is $6 + 12 = 18$ altogether.

8. (a) Suppose a, b, c and d are positive real numbers.

i) Prove that $\frac{a}{b} + \frac{b}{a} \geq 2$.

Solution:

$$\begin{aligned}(a-b)^2 &\geq 0, \\ a^2 - 2ab + b^2 &\geq 0, \\ a^2 + b^2 &\geq 2ab, \\ \therefore \frac{a}{b} + \frac{b}{a} &\geq 2 \text{ as } a, b > 0.\end{aligned}$$

ii) Deduce that $\frac{a+b+c}{d} + \frac{b+c+d}{a} + \frac{c+d+a}{b} + \frac{d+a+b}{c} \geq 12$.

Solution: Similarly

$$\begin{aligned}\frac{a}{c} + \frac{c}{a} &\geq 2, \\ \frac{a}{d} + \frac{d}{a} &\geq 2, \\ \frac{b}{c} + \frac{c}{b} &\geq 2, \\ \frac{b}{d} + \frac{d}{b} &\geq 2, \\ \frac{c}{d} + \frac{d}{c} &\geq 2.\end{aligned}$$

Adding, $\frac{b}{a} + \frac{c}{a} + \frac{d}{a} + \frac{a}{b} + \frac{c}{b} + \frac{d}{b} + \frac{a}{c} + \frac{b}{c} + \frac{d}{c} + \frac{a}{d} + \frac{b}{d} + \frac{c}{d} \geq 12$,

i.e. $\frac{b+c+d}{a} + \frac{a+c+d}{b} + \frac{a+b+d}{c} + \frac{a+b+c}{d} \geq 12$.

iii) Hence prove that if $a + b + c + d = 1$, then:

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d} \geq 16.$$

Solution: Now

$$\begin{aligned}a+b+c &= 1-d, \\ a+b+d &= 1-c, \\ a+c+d &= 1-b, \\ b+c+d &= 1-a,\end{aligned}$$

$$\therefore \frac{1-a}{a} + \frac{1-b}{b} + \frac{1-c}{c} + \frac{1-d}{d} \geq 12,$$

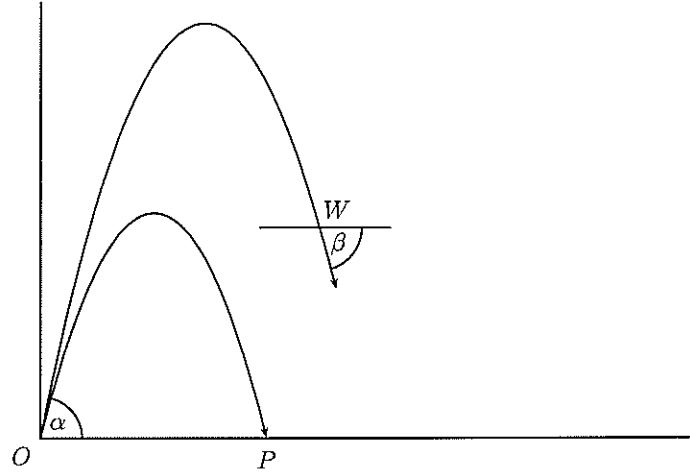
$$\frac{1}{a} - 1 + \frac{1}{b} - 1 + \frac{1}{c} - 1 + \frac{1}{d} - 1 \geq 12,$$

So $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} + \frac{1}{d} \geq 16$.

- (b) Two stones are thrown simultaneously from the same point O in the same direction and with the same non-zero angle of projection α , but with different velocities U and V ($U < V$).

The slower stone hits the ground at a point P on the same level as the point of projection.

At that instant the faster stone is at a point W on its downward path, making an angle β with the horizontal.



- i) Show that $V(\tan \alpha + \tan \beta) = 2U \tan \alpha$.

Solution: For the OP path, For the OW path,
 $\ddot{x} = 0, \quad \ddot{y} = -g, \quad \ddot{x} = 0, \quad \ddot{y} = -g,$
 $\dot{x} = U \cos \alpha, \quad \dot{y} = U \sin \alpha - gt, \quad \dot{x} = V \cos \alpha, \quad \dot{y} = V \sin \alpha - gt,$
 $x = Ut \cos \alpha, \quad y = Ut \sin \alpha - g \frac{t^2}{2}, \quad x = Vt \cos \alpha, \quad y = Vt \sin \alpha - g \frac{t^2}{2}.$
 At P , $t = \frac{2U \sin \alpha}{g}$.
 So at W , $\dot{x} = V \cos \alpha, \quad \dot{y} = V \sin \alpha - 2U \sin \alpha$.
 $-\tan \beta = \frac{\dot{y}}{\dot{x}} = \frac{\sin \alpha}{\cos \alpha} - \frac{2U \sin \alpha}{V \cos \alpha},$
i.e. $-V \tan \beta = V \tan \alpha - 2U \tan \alpha,$
 $V(\tan \alpha + \tan \beta) = 2U \tan \alpha.$

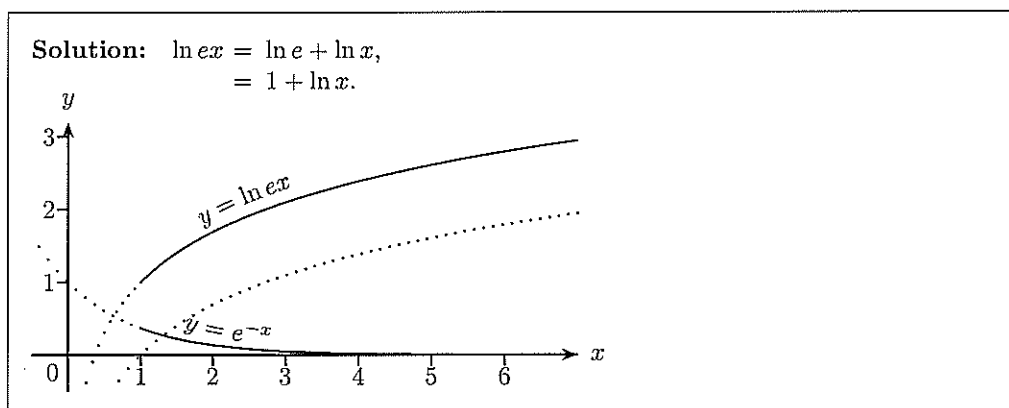
- ii) Deduce that if $\beta = \frac{1}{2}\alpha$, then $U < \frac{3}{4}V$.

Solution: $V \left(\frac{2 \tan \frac{\alpha}{2}}{1 - \tan^2 \frac{\alpha}{2}} + \tan \frac{\alpha}{2} \right) = \frac{2U \times 2 \tan \frac{\alpha}{2}}{1 - \tan^2 \frac{\alpha}{2}},$
 $V \left(2 \tan \frac{\alpha}{2} + \tan \frac{\alpha}{2} - \tan^3 \frac{\alpha}{2} \right) = 4U \tan \frac{\alpha}{2},$
 $V \left(3 - \tan^2 \frac{\alpha}{2} \right) = 4U, \text{ (as } \tan \frac{\alpha}{2} \neq 0)$
 $U = \frac{3V}{4} - \frac{\tan^2 \frac{\alpha}{2}}{4},$
i.e. $U < \frac{3V}{4} \text{ (as } \tan^2 \frac{\alpha}{2} > 0).$

(c) i) Show by graphical means that

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$$\ln ex > e^{-x} \text{ for } x \geq 1.$$



ii) Hence, or otherwise, show that

$$\ln(n!e^n) > e^{-n} \left(\frac{e^n - 1}{e - 1} \right).$$

Solution: $\ln n!e^n = \ln ne.(n-1)e.(n-2)e.(n-3)e \dots (1)e,$
 $= \ln ne + \ln(n-1)e + \ln(n-2)e + \dots + \ln e,$
 $> e^{-n} + e^{1-n} + e^{2-n} + \dots + e^{-1},$
 $> e^{-n}(1 + e + e^2 + \dots + e^{n-1}),$
 $> e^{-n} \left(\frac{e^n - 1}{e - 1} \right).$